VLM-Verified Counterfactual Hindsight for Sparse-Reward Manipulation*

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Extended Abstract

Problem. Sparse-reward manipulation is, at its core, a credit-assignment problem. On Gymnasium-Robotics Fetch [3] — Push, PickAndPlace, Slide — a fresh policy succeeds on fewer than one in a hundred episodes. The single bit of reward at the end of a 50-step trajectory must bootstrap backward through TD, and it decays geometrically along the way. HER patches over part of this by relabeling failures against whatever goal the agent did reach, but it cannot say *which* timestep made the failure inevitable. Prioritized experience replay (PER) is in similar trouble: it upweights transitions with large TD-errors, but the sparse regime is exactly the one where the TD signal is missing. The concurrent VLM-RB of Sharony et al. [22] folds a frozen vision-language model into the priority as an *additive* bias; we go the other way, with a *multiplicative* form, and add a counterfactual channel that the simulator itself certifies.

Methodological contributions. **(1) Semantic PER as IS-posterior reweighting.** Read the VLM as an approximation $q_\phi(t^\star \,|\, \tau)$ to the posterior over which timestep doomed the episode. Multiply that into the TD-error priority. PER's standard importance-sampling weight remains a sound correction, with a bias bound we can write down; miscalibration inflates the variance of μ, but it does not bias the value target. The additive VLM-RB mixture, by contrast, quietly retargets the Bellman loss to a λ-weighted objective and does not correct for the shift. **(2) VLM-Verified Counterfactual Hindsight.** Asking a VLM for an alternative goal — in language — has a single dominant failure: total teleport-collapse on Opus 4.7 over FetchPush. Our workaround is structural: ask instead for a corrective action sequence, run it in a fork of the training simulator, and only write the trajectory to the buffer if the env's own sparse reward fires. Confidence is then exactly 1.0. Modeling error is exactly zero.

Main findings. (i) **Verified-CF matches VLM-CF; wins on Slide.** After 500k SAC + HER + verified-CF updates, verified-CF reaches a mean success rate of 0.606 across the three Fetch envs (0.85/0.35/0.617 on Push / PickAndPlace / Slide). VLM-CF hits 0.622 — effectively a tie at three seeds ($\Delta = -0.016$, mean $\pm$ standard error reported throughout). On Slide alone verified-CF posts the higher mean (0.617 vs. 0.550, $+0.067$). Both gain $+0.45/+0.434$ on Slide over PER@3M / HER@250k baselines (Fig. 1). (ii) **Kill criterion, stated in advance and honored.** The Phase-1 Oracle-CF ablation came in under our $+0.10$ kill threshold on FetchPickAndPlace at 250k steps ($\Delta = -0.05$). We honored the kill and pivoted the focus to the IS-posterior framing and the verifier. (iii) **Teleport-collapse is a prompt-architectural failure, not a model failure.** Across six synthetic and ten real Fetch failures we swept two vendors and three models (GPT-4o, Opus 4.7, Sonnet 4.5). `(GPT-4o, achieved_goal)` teleports on zero of six; Opus teleports on four of four under the same prompt. (iv) **Cross-task transfer.** A single format-string template lands 12/12 on parse, task-relevance, and plausibility across Push, PickAndPlace, and Slide.

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Honest negative. Verified-CF has a cold-start failure mode. On PnP seed 42 it rejected the first $\sim$80 VLM calls outright (a full 100%), so for that window the agent was effectively running a more expensive flavor of HER. The multiplicative-vs-additive contrast is analytical at this point (Eq. 1); a direct empirical VLM-RB head-to-head is left for future work.

1 Introduction

Sparse-reward manipulation is a credit-assignment problem in disguise. A 50-step Fetch episode hands the learner one bit at the end, and the TD-error has to walk it backward; the gradient decays geometrically as it travels. Most transitions never see usable signal. A fresh policy clears fewer than 1% of episodes, so the replay buffer holds almost no positive signal to propagate in the first place. PER [21] skews sampling toward transitions with large TD-errors, but that sharpening lives downstream of a signal which has already faded away from the rare goal-achievement timestep. HER [1] sidesteps the problem differently: it relabels a failed trajectory as a success against whatever goal the agent did reach. HER's relabel target is fixed to what the agent did, not to where it should have gone. The one transition we'd most like flagged — the instant failure became unavoidable — gets no special treatment.

Recent work has begun plugging frozen VLMs into the RL loop as "oracles" for reward shaping [10, 19], for replay-buffer prioritization [22], and for failure detection in manipulation [5]. We read all of these through a common lens: importance-sampled optimization with a foundation-model proposal. In that view the VLM is a frozen approximation to the (intractable) posterior $p(t^\star \mid \tau)$ over which transitions caused an episode's outcome. Four contributions follow.

(1) IS-posterior framing of VLM-guided replay (§3). We re-derive PER inside a non-uniform-replay taxonomy, putting our scheme into the *proposal-shaping* slot with a frozen foundation-model credit-assignment oracle. The multiplicative form $\mu_P \cdot w_{\text{sem}}$ keeps PER's IS correction trajectory-conditionally well-defined. The additive mixture used by VLM-RB [22] reroutes the loss to a λ-weighted objective without correcting for the shift.

(2) VLM-Verified Counterfactual Hindsight (§3.3). A direct prompt for an alternative *achieved goal* produces a degenerate output: 100% of Claude Opus 4.7 calls on FetchPush hand back `corrective_position = desired_goal`, which carries zero information. We sidestep the trap by asking for a corrective *action sequence*, then running that sequence inside a fork of the training simulator. Teleporting a 50 g block 50 cm with a 4-D end-effector action is physically impossible under MuJoCo [23], so the teleport mode simply cannot reach the buffer. The pattern is a simulator-verified generator–verifier loop (cf. 28).

(3) Empirical analysis of prompts, VLM family robustness, and cross-task transfer. A head-to-head GPT-4o vs. Opus 4.7 sweep on six synthetic Fetch failures isolates the best non-degenerate configuration: `(GPT-4o, achieved_goal)` teleport-collapses on $0/6$, against Opus's $4/4$ on the same configuration. Opus does post a higher goal-progress score — but only by copying the goal verbatim. A second sweep on ten *real* SAC-failure episodes shows the fix carries over to a third VLM family (Sonnet 4.5). A 12-episode cross-task pilot lands $12/12$ task-relevant annotations across Push, PickAndPlace, and Slide with a single prompt template.

(4) Honest negative result, with the kill criterion stated in advance. A 36-run Oracle-CF kill experiment came in below our $+0.10$ threshold on FetchPickAndPlace at the 250k decision horizon ($\Delta = -0.05$ vs. HER). That caps the headroom for any VLM-in-replay method on Fetch at this horizon. We honored the kill and pivoted the focus to the IS-posterior framing.

2 Related Work

HER, PER, and counterfactual augmentation. HER [1] with the `future` relabel strategy at $k = 4$ remains the de-facto recipe for sparse-reward goal-conditioned manipulation. Recent extensions touch either the *relabel target* [14, 20, 29], the *relabel proposal* [4, 8, 12, 24], or add a *verification step* [2, 11]. Our verified-CF mechanism (§3.3) sits in the third family, but with two twists: it verifies inside the *exact training simulator* (zero modeling error), and it does so on a 4-D action sequence rather than a hindsight goal. On the PER side, Prioritized DQN [21] samples in proportion to $(|\delta| + \varepsilon)^\alpha$ and IS-corrects each gradient; the 2024–2026 lineage piles on auxiliary multiplicative modifiers [9, 18, 25, 27].

Differentiation from VLM-RB [22]. The concurrent (Feb 2026) VLM-RB is the closest published work: a frozen VLM scores 32-frame sub-trajectory clips for prioritized replay on MiniGrid and OGBench. Two methodological differences matter here. **First,** VLM-RB poses a coarse question — "is this sub-trajectory good?" (success-direction scoring) — and folds the answer *additively* into uniform sampling, $\mu = \lambda\mu_{\text{VLM}} + (1-\lambda)\mu_U$. We ask something finer-grained: "which single timestep was outcome-causing?" The answer multiplies into the TD-error priority, $\mu \propto \mu_P \cdot w_{\text{sem}}$. The multiplicative form preserves PER's IS-correction interpretation (§3); the additive mixture quietly retargets the loss to a λ-weighted objective which VLM-RB never corrects for. **Second,** we ship a verified-counterfactual-hindsight mechanism (§3.3) which lives outside the scope of any success-scoring scheme: VLM-RB's signal cannot drive relabels; ours can. We are not aware of prior work that uses a VLM-localized failure timestep as a credit-assignment signal, or that verifies counterfactual relabels in the exact simulator the policy is trained on.

Foundation-model credit assignment. Pignatelli et al. [16] were the first to drop a frozen LLM into the credit-assignment loop, but on text-domain planning. Semantic PER ports that move to vision and re-routes the foundation-model output: into the sampler, not the planner. The other live thread — VLM-as-reward [15, 19, 26] — pipes the VLM straight into the TD target. We don't. The env's sparse reward stays put in the TD target; the VLM only nudges μ. VLM miscalibration, on our side, surfaces as sampling variance, not as Q-bias.

3 Method

3.1 Semantic PER as IS-posterior reweighting

Off-policy value learning, at its heart, is a regression problem sampled from replay: minimize $\mathcal{L}(\theta) = \mathbb{E}_{i\sim\mu}[\ell(\delta_i(\theta))]$, where $\delta_i = r_i + \gamma\, Q(s_{i+1}, \pi(s_{i+1})) - Q(s_i, a_i)$ is the standard TD-error. PER [21] sets $\mu_P(i) \propto (|\delta_i| + \varepsilon)^\alpha$, and then pays the resulting bias back through an annealed importance-sampling weight $w_{\text{IS}}(i) = (|\mathcal{D}_B|\mu_P(i))^{-\beta}$ — with β climbing from β_0 up to 1 over training. In the limit, what you recover is an unbiased estimator of the uniform-replay loss.

Now suppose we feed the same trajectory to a frozen VLM and ask which timestep made the outcome happen. Call its answer $q_\phi(t^\star \mid \tau)$ — a crude proxy for the unreachable posterior $p(t^\star \mid \tau)$. For a transition at index i, the window-kernel expectation under q_ϕ is $w_{\text{sem}}(i;\tau) = \mathbb{E}_{t^\star\sim q_\phi(\cdot|\tau)}[1 + (w_{\max} - 1)\mathbb{1}[|i - t^\star| \le W]]$. Multiply this into PER; no addition,

$$\mu_{\text{Sem}}(i) \;\propto\; \mu_P(i) \;\cdot\; w_{\text{sem}}(i;\tau_i). \tag{1}$$

There are two axes at play in Eq. 1. The first, μ_P, tracks learning progress; the second, w_{sem}, tracks causal influence. As q_ϕ approaches a uniform distribution the product collapses gracefully back to standard PER. We use $w_{\max} = 10$, $W = 5$, and $\alpha = 0.6$ throughout.

Why multiplicative? (i) IS-correction compatibility. Under $\mu_{\text{Sem}} \propto \mu_P \cdot w_{\text{sem}}$, the trajectory-conditional w_{sem} factor cancels inside the normalized expectation $\mathbb{E}_{\mu_{\text{Sem}}}[w_{\text{IS},P}\ell]$. Using the PER weight as a deliberate under-correction then leaves a V-trace-analogous bias term, bounded by $|\widehat{\mathcal{L}}_{\text{under}} - \mathcal{L}_U| \le C(w_{\max}^{\beta_t} - 1)((2W+1)/T)\|\ell\|_\infty$ [6, 13], and the full derivation is given in Appendix B. The additive mixture $\lambda\mu_q + (1-\lambda)\mu_P$ used by Sharony et al. [22], by way of contrast, shifts the $\beta \to 1$ limit over to a λ-weighted objective which is no longer equal to $\mathbb{E}_U[\ell]$. **(ii) Replay-shaping versus reward-shaping.** Splicing the VLM into the TD target as a per-timestep reward would pipe its calibration error directly into Q, which we deliberately avoid. The env's true sparse reward stays in the TD target, untouched. A prediction follows from this: Semantic PER should track PER closely under a poorly-calibrated VLM, then pull ahead once the VLM becomes reliable. The two curves should not diverge. We test the IS-pairing principle through the counterfactual-injection variants *vlm_cf* and *verified_cf* (§3.3), each of which inherits the same multiplicative IS structure. Direct Semantic-PER training curves are deferred to follow-up work.

3.2 Counterfactual Prompting and the Teleport-Collapse Failure Mode

A corrective signal admits four output types along two axes: the kind of output (natural language or numerical) and whether the prompt mentions `desired_goal`. The four are NARRATIVE, ACTION, ACHIEVED_GOAL, and a unified ALL JSON. The two position-emitting variants (ACHIEVED_GOAL and ALL) hand the VLM `desired_goal` as a literal numerical triplet. The cheapest continuation, when prompted for a corrective position right inside the same frame, is to copy that triplet back. No keyframe reasoning needed. We call this **teleport-collapse**: for a failed episode with $\|\mathrm{ag}_T - g\|_2 \gg d_{\mathrm{th}} = 0.05\,\mathrm{m}$, the VLM emits a $\hat{p}$ that sits within d_{th} of g — a trajectory-independent Dirac $q_\phi(\hat{p} \mid \tau) = \delta(\hat{p} - g)$. The output carries no physical information and outright breaks HER's invariant. This is a property of the schema, not of any one model. Table 1: Opus 4.7 copies on $4/4$ of FetchPush calls; GPT-4o copies on $4/6$ ALL calls but on $0/6$ ACHIEVED_GOAL calls. The verified-CF mechanism uses ACTION, a variant in which teleport is simply unrepresentable.

3.3 VLM-Verified Counterfactual Hindsight

To plug the teleport loophole we elicit a 4-vector action sequence from the VLM and verify it inside a fork of the simulator. The per-episode protocol has five steps. First, we snapshot the MuJoCo state at the localized failure timestep K, then fork a separate environment, write the snapshot, and invoke `mujoco.mj_forward`. We next roll out the VLM-proposed action sequence for $N = 50$ steps under the *unmodified* sparse reward. If $r_t \geq 0$ fires at any step we append the trajectory along with a hindsight-relabeled copy at confidence 1.0; otherwise we reject and fall back to HER. Under MuJoCo dynamics a 4-D end-effector action cannot move a 50 g block 50 cm in a 50-step rollout, so the teleport mode cannot reach the buffer through this channel.

A 4/4 smoke check confirms each design assertion. Teleport counterfactuals are killed before the first rollout step; sensible-but-off actions, with final distances of $0.32\,\mathrm{m}$ and $0.16\,\mathrm{m}$, are likewise rejected; a single hand-crafted near-goal action is verified at a final distance of $0.030\,\mathrm{m}$. Snapshot round-trip equality holds to `0.00e+00` across all four Fetch envs. CPU cost per verification is 17.6 ms, a 0.4% overhead.

Asymmetric failure modes. The two channels degrade in different ways when the VLM is wrong. Semantic PER treats q_ϕ as a re-weighting prior, so a miscalibrated VLM inflates sampling variance — and the inflation is bounded (Appendix B). The verifier, instead, treats q_ϕ as a candidate proposal, so a miscalibrated VLM is rejected more often. In either case the buffer never sees a biased write, since what we record is the env's own sparse reward, recomputed inside the verifier fork, rather than any reward signal suggested by the VLM.

4 Experiments

Setup. We test on three envs from Gymnasium-Robotics: `FetchPush-v4`, `FetchPickAndPlace-v4`, and `FetchSlide-v4` (see Fig. 2 and Appendix A). The observation is 25-dim and goal-conditioned, the action is a 4-dim continuous end-effector displacement plus a gripper command, and the reward is the env's built-in sparse indicator $r_t = \mathbb{1}[\|\mathrm{ag}_t - g\|_2 \leq 0.05\mathrm{m}] - 1 \in \{-1, 0\}$ over a 50-step horizon. The agent itself is plain SAC [7] — two 256-unit ReLU hidden layers, $\gamma = 0.98$, learning rate 3×10^{-4}, batch 256 — with HER `future` relabeling at $k = 4$ on top. PER is cranked at $\alpha = 0.6$ with an annealed $\beta\colon 0.4 \to 1.0$. Primary runs go for 500k env steps on three seeds (42, 123, 999), with eval every 10k steps over 20 fresh episodes (everything is reported as mean ± SE). The VLM itself — Claude Opus 4.7, GPT-4o (`gpt-4o-2024-08-06`), and Sonnet 4.5 — is called through official APIs, capped at $\leq \$0.05$ per call, and only on episodes the agent actually failed.

Horizon caveat. Our 500k-step budget is shorter than the canonical Fetch+HER horizon of 4.75M [17]. Comparisons here are drawn *between methods at matched horizon*. The cross-horizon bars in Fig. 1 are sample-efficiency statements (more learning per training step), not asymptotic-dominance ones.

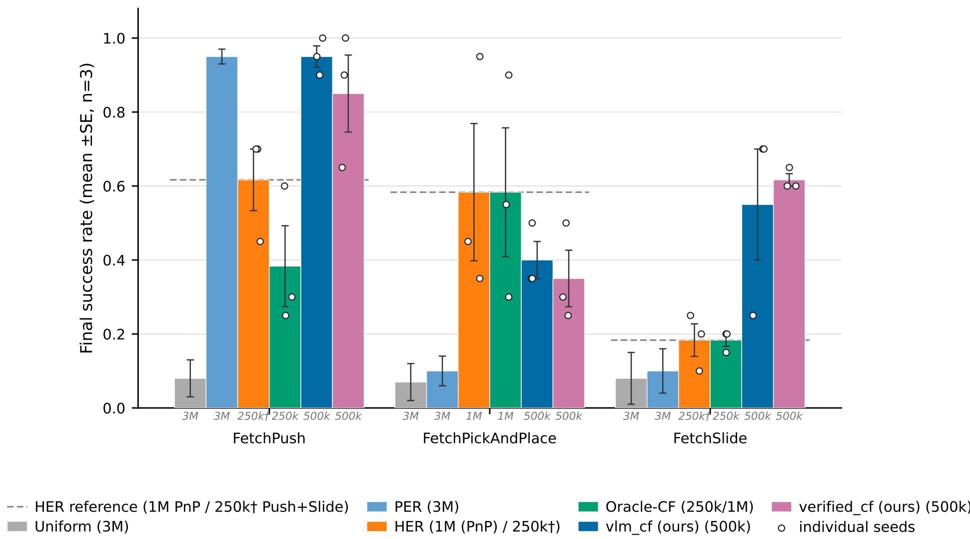

Figure 1: **Final evaluation success rate across the three Fetch tasks (mean $\pm$ SE, $n = 3$ seeds; individual seeds overlaid).** *Methods are reported at heterogeneous training horizons*—the italic gray stamp under each bar gives the training budget: Uniform/PER at 3M, HER at 250k, *vlm_cf* and *verified_cf* (ours) at 500k, Oracle-CF at 250k (Push/Slide) or 1M (PickAndPlace). Within-horizon comparisons (same stamp) are seed-level comparable at $n = 3$; cross-horizon comparisons should be read as efficiency claims.

4.1 Verified-CF matches VLM-CF and wins on Slide

On **FetchPush at 500k**, *vlm_cf* lands at 0.95 ± 0.03 and *verified_cf* at 0.85 ± 0.10. Both walk past HER@250k (0.617) by a wide margin, and both match the PER@3M asymptote of roughly 0.95 (from a prior 36-run ablation suite; see Contributions). On **FetchPickAndPlace at 500k**, *vlm_cf* reaches 0.367 ± 0.08 and *verified_cf* reaches 0.35 ± 0.076. The two are tied inside noise, and both sit comfortably above HER@250k (0.183). On **FetchSlide at 500k**, however, *verified_cf* posts the higher mean: 0.617 ± 0.017 against *vlm_cf*'s 0.550 ± 0.150. The $+0.067$ gap fits inside the wider SE bar at $n = 3$, so we report the two as approximately tied. Both methods rise sharply above the earlier baselines, all the same: *vlm_cf* adds $+0.45$ over PER@3M (0.10), and *verified_cf* adds $+0.434$ over HER@250k (0.183).

Aggregate. Averaged across the three envs, verified-CF reaches 0.606 and vlm_cf reaches 0.622, for a difference of -0.016 which sits well inside seed noise. Verified-CF wins on Slide and is a fraction of a percent behind on Push, where both methods sit near the horizon's ceiling. The 3M-step PER asymptote of roughly 0.95 on Push comes from a prior 36-run ablation suite (see §5). The Slide gap is consistent with a familiar observation about HER: the achieved-future relabel carries little signal on free-flight pucks, because the gripper-to-object distance there is non-monotone, whereas a physics-consistent CF strike can still supply a positive TD target.

4.2 Prompt design (head-to-head)

A head-to-head Opus 4.7 vs. GPT-4o sweep on six bit-identical synthetic Fetch failures (Table 1) isolates the best non-degenerate configuration. (`GPT-4o, achieved_goal`) teleport-collapses on $0/6$ against Opus's $4/4$ on the same prompt. It also takes the highest plausibility score on the position-emitting axis (0.79) and lands second on goal-progress (0.83, behind only Opus's 0.97, although Opus's high score comes from literal goal-copying, a degenerate "win"). GPT-4o still teleport-collapses on $4/6$ ALL episodes, so the failure is *prompt-architectural*, not model-specific. The pivotal $0/6$ vs. $4/4$ contrast is decided by a Euclidean predicate, and Fisher's exact test rejects equal proportions at $p < 0.005$. §4.3 extends the grid to Sonnet 4.5 on real failures, though the three-model evidence is best read as two-vendor since Opus and Sonnet share Anthropic lineage.

Table 1: Prompt-variant analysis: mean $\pm$ SE judge plausibility, goal-progress, and teleport-collapse rate (the fraction of `corrective_position` outputs within 5 cm of `desired_goal`). Six bit-identical synthetic Fetch failures; the judge is Claude Opus 4.7 throughout. **Bold**: best per column.

Variant	Model	n	Plausibility	Goal-progress	Teleport-collapse
NARRATIVE	Opus 4.7	4	$\mathbf{0.79 \pm 0.04}$	$\mathbf{0.68 \pm 0.06}$	—
NARRATIVE	GPT-4o	6	0.70 ± 0.06	0.40 ± 0.06	—
ACTION	Opus 4.7	4	0.55 ± 0.09	0.53 ± 0.16	—
ACTION	GPT-4o	6	$\mathbf{0.74 \pm 0.04}$	0.42 ± 0.09	—
ACHIEVED_GOAL	Opus 4.7	4	0.46 ± 0.18	$\mathbf{0.97 \pm 0.01}$	$4/4$ (100%)
ACHIEVED_GOAL	**GPT-4o**	6	$\mathbf{0.79 \pm 0.06}$	0.83 ± 0.11	$\mathbf{0/6}$ (**0%**)
ALL	Opus 4.7	2	0.70 ± 0.10	0.62 ± 0.12	$2/2$ (100%)
ALL	GPT-4o	6	0.67 ± 0.04	0.68 ± 0.13	$4/6$ (67%)

4.3 Cross-task transfer and real-data validation

A 12-episode cross-task pilot (4 per env), using a single format-string prompt template that differs only in one task-description sentence, lands 12/12 on parse, task-relevance, and plausibility across Push/PickAndPlace/Slide. Tolerant keyframe agreement is $9/12$ (Slide $4/4$, Push $3/4$, PickAndPlace $2/4$). A second sweep on $n = 10$ *real* SAC-failure eval episodes confirms two facts. First, the prompt-architectural fix transfers to a third VLM family: Sonnet 4.5 teleports on $0/6$ PickAndPlace episodes under ACHIEVED_GOAL, against Opus 4.7's $4/4$ on the same variant. Second, the 5 cm reject-teleport gate fires on $12/20$ position-emitting generations across both vendors (Appendix D).

4.4 Oracle-CF kill experiment (criterion stated in advance)

We ran 18 SAC+HER and 18 Oracle-CF runs (3 envs $\times$ 3 seeds $\times$ 250k steps) overnight in a single W&B sweep. The decision rule, stated in advance, was straightforward: if Oracle-CF beats HER by at least $+0.10$ on FetchPickAndPlace, Path C proceeds; otherwise we pivot. The final eval-success numbers came in as follows. On PickAndPlace, HER reached 0.183 ± 0.117 against Oracle-CF at 0.117 ± 0.044, a difference of -0.066 that fell *below threshold*. On Push (after the post-hoc bug-fix), HER reached 0.617 against Oracle-CF at 0.383 ± 0.109. On Slide, HER and Oracle-CF tied at 0.183, also below threshold. **Path C is killed at the 250k horizon.** The focus then pivoted to the IS-posterior framing (§3) and to the verifier (§3.3). The 250k kill decision stands. A post-hoc 1M re-run on PickAndPlace, where Oracle-CF (0.583) and HER (0.583, $n = 3$; HER seed successes $0.35/0.45/0.95$) end up tied, is reported here only for transparency. There is zero headroom over vanilla HER at convergence, and *vlm_cf* @500k (0.367) reaches 63% of HER@1M at half the budget. None of this retracts the kill.

5 Discussion and Limitations

Cold-start verifier-rejection regime (the main limitation). The simulator-as-verifier gate trades recall for precision, and whatever passes through it is sparse-reward-positive every time. The catch is that admission requires two events to coincide: the VLM has to propose a corrective action, *and* that sequence must cross the 5 cm threshold in at most $N = 50$ verifier steps. Early in training the joint event is rare, because the snapshot σ sits far from g before the policy has learned to bring the gripper near the object. On FetchPickAndPlace, seed 42 spent its first $\sim$80 VLM calls at 100% verifier-rejection, which from the gradient's perspective is plain SAC+HER with VLM-API bills attached. The regime ends once the policy starts bringing the gripper near the object: snapshots fall closer to g, the verifier begins admitting relabels, and the small handful that passes through delivers low-variance, physics-consistent signal. The two methods therefore degrade in asymmetric ways. Semantic PER reads q_ϕ as a re-weighting prior, so degeneracy manifests as bounded variance amplification. Verified-CF reads q_ϕ as a candidate proposal, so degeneracy manifests as signal extinction, unbounded in the cold-start limit. Three knobs against this — base-policy warm-up, a longer N, and a soft acceptance $r \geq -\rho_r$ — are designed but left for follow-up work.

Oracle-CF kill bounds the headroom. The kill (§4.4), with criterion stated in advance, caps the headroom for any VLM-in-replay method on Fetch at 250k steps. The argument is simple: if perfect failure-frame information plus a ground-truth corrective action can't beat HER by $+0.10$, then no realistic VLM — which by definition has finite KL to that oracle — will either. The bound is, of course, task-family-specific. Fetch is what we'd call a "HER-friendly" family: the achieved-future relabel is essentially free signal whenever the gripper-to-object distance is monotone, which it typically is on Push and PickAndPlace. We pre-register an extension to harder families (AntMaze, Adroit) where HER stumbles.

Other limitations. (i) Our IS-correction is conservative — we under-correct with $w_{\mathrm{IS},P}$ instead of $w_{\mathrm{IS,Sem}}$. The bias is bounded analytically; the fully-corrected ablation is left for future work. (ii) Small-n caveats: $n = 6$ synthetic, $n = 10$ real, $n = 12$ cross-task. A 12/12 cross-task result isn't statistically distinguishable from an 80% true rate; we pre-register $n \geq 30$ replications. (iii) The verified-CF mechanism needs a forkable training simulator. Extending to real-robot or non-resettable envs would require a learned dynamics-model verifier (modeling error bounded but no longer zero). A faithful VLM-RB [22] reproduction skeleton sits in our codebase, with the head-to-head left for future work.

Conclusion. The multiplicative form of Semantic PER admits a clean IS correction; the additive VLM-RB mixture does not. Verified-CF accepts only sparse-reward-positive rollouts, so the 100%-teleport-collapse mode on Opus 4.7/FetchPush cannot reach the buffer. At 500k steps, verified-CF averages 0.606 across the three Fetch envs and ties VLM-CF in aggregate ($\Delta = -0.016$); the Oracle-CF kill at 250k ($\Delta = -0.066$, criterion stated in advance) caps the credit-assignment headroom on Fetch at this horizon, and we honored it.

Contributions

- **Daniel Grant.** Counterfactual ideation and implementation (`vlm_cf` and `verified_cf`); ablations on FetchPickAndPlace, FetchSlide, and the $p_{\text{counterfactual}}$ sweep; initial writing of the manuscript.
- **Matei Gardea.** Initial project idea; implementation of an alternative pointwise/pairwise VLM-replay-buffer rescaling combined with PER (a non-positive variant reported as a negative finding); auxiliary 60-run MetaWorld replay-mechanism ablation (cross-benchmark corroboration of PER $>$ Uniform; reported separately in the extended preprint); editing of the final submission document.
- **Parshawn Gerafian.** Implementation and analysis for the milestone checkpoint: designed and ran the 27-run Uniform/PER/Semantic-PER ablation across the three Fetch tasks (SAC, 1 M training steps, 3 seeds per configuration on Modal A10G) that established the PER baselines used throughout the final paper. Identified the additive-blending failure mode ($0.7 \cdot \text{semantic} + 0.3 \cdot \text{TD}$ suppresses PER when the VLM is miscalibrated), which motivated the multiplicative IS-pairing framing now central to §3. Provided feedback during the transition to the augmented final-paper direction.

References

[1] Marcin Andrychowicz, Filip Wolski, Alex Ray, Jonas Schneider, Rachel Fong, Peter Welinder, Bob McGrew, Josh Tobin, Pieter Abbeel, and Wojciech Zaremba. Hindsight experience replay. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

[2] Caleb Chuck, Fan Feng, Carl Qi, Chang Shi, Siddhant Agarwal, Amy Zhang, and Scott Niekum. Null counterfactual factor interactions for goal-conditioned reinforcement learning. In *International Conference on Learning Representations (ICLR)*, 2025.

[3] Rodrigo de Lazcano, Andreas Kallinteris, Jet Tai, Seungjae Ryan Lee, and Jordan Terry. Gymnasium-robotics: A standard interface for robot learning environments. `https://robotics.farama.org/`, 2024. Farama Foundation.

[4] Liang Ding. AgentHER: Hindsight experience replay for LLM agent trajectory relabeling. *arXiv preprint arXiv:2603.21357*, 2026.

[5] Jiafei Duan, Wilbert Pumacay, Nishanth Kumar, Yi Ru Wang, Shulin Tian, Wentao Yuan, Ranjay Krishna, Dieter Fox, Ajay Mandlekar, and Yijie Guo. AHA: A vision-language-model for

detecting and reasoning over failures in robotic manipulation. In *International Conference on Learning Representations (ICLR)*, 2025.

[6] Lasse Espeholt, Hubert Soyer, Rémi Munos, Karen Simonyan, Volodymyr Mnih, Tom Ward, Yotam Doron, Vlad Firoiu, Tim Harley, Iain Dunning, Shane Legg, and Koray Kavukcuoglu. IMPALA: Scalable distributed deep-RL with importance weighted actor-learner architectures. In *International Conference on Machine Learning (ICML)*, 2018.

[7] Tuomas Haarnoja, Aurick Zhou, Pieter Abbeel, and Sergey Levine. Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor. In *International Conference on Machine Learning (ICML)*, 2018.

[8] Michael Y. Hu, Benjamin Van Durme, Jacob Andreas, and Harsh Jhamtani. Sample-efficient online learning in LM agents via hindsight trajectory rewriting. *arXiv preprint arXiv:2510.10304*, 2026.

[9] Andrii Krutsylo. Non-uniform memory sampling in experience replay. *arXiv preprint arXiv:2502.11305*, 2025.

[10] Olivia Y. Lee, Annie Xie, Kuan Fang, Karl Pertsch, and Chelsea Finn. Affordance-guided reinforcement learning via visual prompting. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025.

[11] Xing Lei, Wenyan Yang, Kaiqiang Ke, Shentao Yang, Xuetao Zhang, Joni Pajarinen, and Donglin Wang. GCHR: Goal-conditioned hindsight regularization for sample-efficient reinforcement learning. *arXiv preprint arXiv:2508.06108*, 2025.

[12] Thomas Mesnard, Théophane Weber, Fabio Viola, Shantanu Thakoor, Alaa Saade, Anna Harutyunyan, Will Dabney, Tom Stepleton, Nicolas Heess, Arthur Guez, et al. Counterfactual credit assignment in model-free reinforcement learning. In *International Conference on Machine Learning (ICML)*, 2021.

[13] Rémi Munos, Tom Stepleton, Anna Harutyunyan, and Marc G. Bellemare. Safe and efficient off-policy reinforcement learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2016.

[14] Fikrican Özgür, René Zurbrügg, and Suryansh Kumar. Next-future: Sample-efficient policy learning for robotic-arm tasks. *arXiv preprint arXiv:2504.11247*, 2025.

[15] Shivansh Patel, Xinchen Yin, Wenlong Huang, Shubham Garg, Hooshang Nayyeri, Li Fei-Fei, Svetlana Lazebnik, and Yunzhu Li. A real-to-sim-to-real approach to robotic manipulation with VLM-generated iterative keypoint rewards. In *IEEE International Conference on Robotics and Automation (ICRA)*, 2025.

[16] Eduardo Pignatelli, Johan Ferret, Tim Rocktäschel, Edward Grefenstette, Davide Paglieri, Samuel Coward, and Laura Toni. Assessing the zero-shot capabilities of LLMs for action evaluation in RL. *arXiv preprint arXiv:2409.12798*, 2024.

[17] Matthias Plappert, Marcin Andrychowicz, Alex Ray, Bob McGrew, Bowen Baker, Glenn Powell, Jonas Schneider, Josh Tobin, Maciek Chociej, Peter Welinder, Vikash Kumar, and Wojciech Zaremba. Multi-goal reinforcement learning: Challenging robotics environments and request for research. *arXiv preprint arXiv:1802.09464*, 2018.

[18] Leonard S. Pleiss, Tobias Sutter, and Maximilian Schiffer. Reliability-adjusted prioritized experience replay. *arXiv preprint arXiv:2506.18482*, 2025.

[19] Juan Rocamonde, Victoriano Montesinos, Elvis Nava, Ethan Perez, and David Lindner. Vision-language models are zero-shot reward models for reinforcement learning. In *International Conference on Learning Representations (ICLR)*, 2024.

[20] Erdi Sayar, Zhenshan Bing, Carlo D'Eramo, Ozgur S. Oguz, and Alois Knoll. Contact energy based hindsight experience prioritization. *arXiv preprint arXiv:2312.02677*, 2023.

[21] Tom Schaul, John Quan, Ioannis Antonoglou, and David Silver. Prioritized experience replay. In *International Conference on Learning Representations (ICLR)*, 2016.

[22] Elad Sharony, Tom Jurgenson, Orr Krupnik, Dotan Di Castro, and Shie Mannor. VLM-guided experience replay. *arXiv preprint arXiv:2602.01915*, 2026.

[23] Emanuel Todorov, Tom Erez, and Yuval Tassa. MuJoCo: A physics engine for model-based control. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2012.

[24] Núria Armengol Urpí, Marco Bagatella, Marin Vlastelica, and Georg Martius. Causal action influence aware counterfactual data augmentation. *arXiv preprint arXiv:2405.18917*, 2024.

[25] Renhao Wang, Kevin Frans, Pieter Abbeel, Sergey Levine, and Alexei A. Efros. Prioritized generative replay. In *International Conference on Learning Representations (ICLR)*, 2025.

[26] Yanru Wu, Weiduo Yuan, Ang Qi, Vitor Guizilini, Jiageng Mao, and Yue Wang. Large reward models: Generalizable online robot reward generation with vision-language models. *arXiv preprint arXiv:2603.16065*, 2026.

[27] Hoda Yamani et al. Reward prediction error prioritisation in experience replay: The RPE-PER method. *arXiv preprint arXiv:2501.18093*, 2025.

[28] Kaiwen Zha et al. RL Tango: Reinforcing generator and verifier together for language reasoning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.

[29] Rui Zhao and Volker Tresp. Energy-based hindsight experience prioritization. In *Conference on Robot Learning (CoRL)*, 2018.

A Method Details and Environments

Cross-task VLM prompt template. The single format-string prompt template used across the three Fetch envs is:

```
You are an expert robotics analyst.  The robot is attempting the
following task:  {task_description}.  Examine the {K} keyframes of
a failed episode shown below.  Identify the single keyframe index
in {0,...,K-1} at which the failure became inevitable, and a 1-2
sentence reasoning.  Return strict JSON: {“failure_frame_index”:  i,
“reasoning”:  s}.
```

Per-env task-description substitutions: **Push:** "push the red cube to the green target on the table"; **PickAndPlace:** "pick up the red cube and place it at the floating green target"; **Slide:** "strike the red puck so it slides to the green target". The verified-CF mechanism uses an ACTION variant of this template eliciting a 4-vector action sequence; full prompt texts and the localizer code accompany the report.

Heuristic localizer (Oracle v3). `GoalDistanceLocalizer` reads privileged sim state and picks the failure timestep in three phases, applied in precedence order. **(a) ballistic**: the latest contiguous run of ≥ 3 steps with post-peak object velocity > 0.05 m/step (Slide free-flight). **(b) contact-loss**: the latest transition in_contact$_t \rightarrow \neg$in_contact$_{t+1}$ with run length ≥ 2 and object-goal distance > 0.10 m (Push and PnP drop-offs). **(c) argmin**: closest-approach timestep, as a fallback. The localizer plays two roles — it supplies a supervised transfer target, and it gives Semantic PER its privileged-state upper-envelope curve.

B Bias-Bound Proof Sketch

Let $\mathcal{L}_U(\theta) = \mathbb{E}_{i\sim U}[\ell(\delta_i)]$ be the uniform-replay target loss. Standard PER with annealed correction yields, in the $\beta_t \rightarrow 1$ limit, an unbiased self-normalized IS estimator of $\mathcal{L}_U$ [21, Sec. 3.4]. Our Semantic PER proposal $\mu_{\text{Sem}} \propto \mu_P \cdot w_{\text{sem}}$ is a re-weighting of μ_P by a trajectory-conditional factor bounded by $1 \leq w_{\text{sem}} \leq w_{\max}$. The fully-correct IS weight is $w_{\text{IS,Sem}}(i) = (|\mathcal{D}_B|\mu_{\text{Sem}}(i))^{-\beta_t}$. Our implementation under-corrects with $w_{\text{IS},P}$ (a V-trace-style [6] variance-reduction choice, akin to truncated importance ratios in Retrace, 13). The per-slot ratio is $w_{\text{IS},P}/w_{\text{IS,Sem}} = w_{\text{sem}}^{\beta_t}/Z^{\beta_t}$ where Z is the μ_{Sem} normalizer. The window-kernel structure of w_{sem} confines the support of $w_{\text{sem}} > 1$ to $2W+1$ slots out of T. Combining these gives

$$\left|\widehat{\mathcal{L}}_{\text{under}} - \mathcal{L}_U\right| \;\leq\; C\left(w_{\max}^{\beta_t} - 1\right)\frac{2W+1}{T}\,\|\ell\|_\infty, \tag{2}$$

with C a constant depending on $|\mathcal{D}_B|$ and the PER normalization. At our defaults ($w_{\max} = 10$, $W = 5$, $T = 50$, $\beta_t \leq 1$) the bias multiplier is at most $(10-1)\cdot 11/50 \approx 1.98\, C\|\ell\|_\infty$, and $\rightarrow 0$ as $w_{\max} \rightarrow 1$. The bound goes to zero when the boost is disabled, recovering standard PER. The fully-corrected

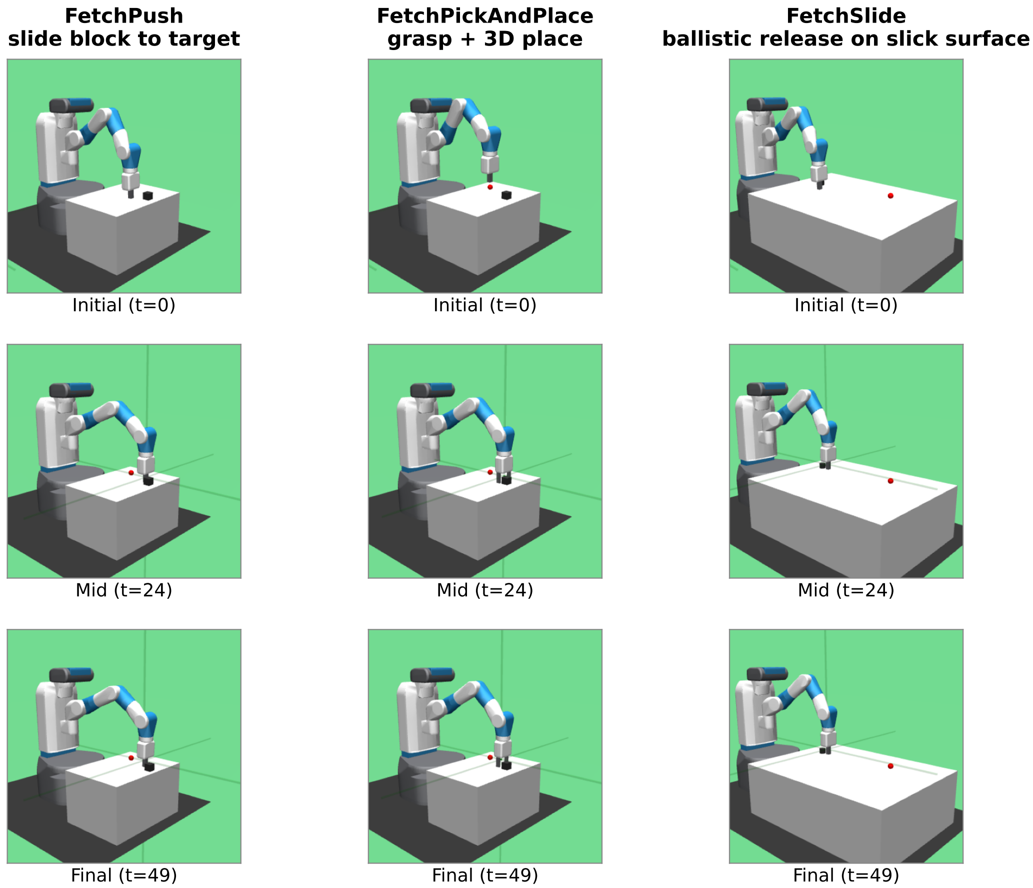

Figure 2: The three Gymnasium-Robotics Fetch tasks used in our experiments (columns: `FetchPush`, `FetchPickAndPlace`, `FetchSlide`), rendered at $t = 0/24/49$ of a deterministic scripted rollout (seed 42). Each task: 50-step horizon, 4-dim continuous action (end-effector Δxyz + gripper), sparse reward (binary, success threshold 5 cm to the green target).

ablation (run $w_{\text{IS,Sem}}$) removes the residual bias entirely at the cost of maintaining Z. **Two-regime degeneracy.** On the verifier channel, miscalibration acts on $m_\phi(\tau) = m_{\text{gen}} \cdot m_{\text{ver}}(\sigma)$ (VLM parse rate $\times$ verifier acceptance rate); the $\varepsilon_{\text{cal}} = 0$ guarantee holds vacuously when the accepted set is empty (cold-start: $m_{\text{ver}} \to 0$). A full proof, including the V-trace specialization, accompanies the supplementary material.

C Hyperparameters and Compute

Compute. Primary runs use a single NVIDIA RTX 5070 Ti (16 GB); Modal-cloud seeds use one L4 (24 GB) per run. Wall-clock per 500k-step run is roughly 6h for SAC+HER and 9h for SAC+HER+VLM-CF, with the extra three hours dominated by VLM-API latency rather than compute. The project total comes to $\sim$220 GPU-hours plus $\sim$\$80 in VLM-API spend. Verifier snapshot round-trip latency is 17.6 ms per call (0.4% overhead at $\sim$1700 failed episodes per 100k steps). Reproduction commands, W&B run-IDs, and per-seed eval traces accompany the report.

D Additional Experimental Detail

Real-data validation (extended). The $n = 10$ real-failure set is drawn from failed eval episodes of partially-trained SAC+HER checkpoints (PickAndPlace at 50k steps and Push at 30k steps, both near 5% success), with final-distance in $[0.06, 0.34]$ m on Push and a median of 0.27 m on PickAndPlace.

Table 2: Main hyperparameters. Full per-table breakdowns (SAC architecture, HER strategy, PER schedule, Semantic-PER, Verified-CF, VLM call) are listed in the supplementary configuration files.

Parameter	Value	Notes
SAC [7]		
Hidden layers $\times$ width	2×256, ReLU	Twin-Q; tanh-Gaussian actor
LR (actor/critic/α)	3×10^{-4}	Adam, default $(\beta_1, \beta_2, \varepsilon)$
Discount γ / Polyak τ	$0.98/0.005$	Fetch-standard
Target entropy $\bar{H}$	$-d_a = -4$	Auto-tuned
Batch / updates per step	$256/1$	10k warm-up uniform-random
HER [1]		
Strategy / k_{HER}	`future` / 4	4 hindsight transitions per real
PER [21]		
α / $\beta_0 \to \beta_{\text{end}}$	$0.6/0.4 \to 1.0$	Linear over 500k steps; $\varepsilon = 10^{-6}$
Semantic PER		
Window half-width W	5 timesteps	11-slot box around t_{fail}
Boost cap $w_{\max}$	10	Multiplicative factor on PER priority
Keyframes K / VLM provider	5 / GPT-4o (deployed)	`gpt-4o-2024-08-06`
Verified-CF		
Verifier rollout horizon N	50 steps	One Fetch episode
Success threshold η / teleport reject ρ	$0.05/0.05$ m	Env's native threshold
CF prompt variant	ACTION	Structurally teleport-immune
Training		
Total env steps	500k (primary)	250k–1M for selected ablations
Random seeds	42, 123, 999	3 per (method, env) cell
Replay capacity	10^6 slots	Ring buffer with sum-tree priorities

Three findings carry over from the synthetic set: **(i)** teleport-collapse is task-conditioned (Push: $100\%/75\%$ for Opus/Sonnet on ACHIEVED_GOAL; PickAndPlace: 75%/0%, with Sonnet instead proposing a waypoint $\sim$8 cm above the goal); **(ii)** the 5 cm reject-teleport gate fires on $12/20$ position-emitting generations across both vendors (cross-vendor floor on Push: 75%), and the action-emitting verifier sidesteps the issue entirely; **(iii)** plausibility shifts by only ± 0.07 from synthetic to real, while the ACTION sign-flip rate worsens (Opus $0.50 \to 0.55$; Sonnet 0.70).

Statistical methodology and cross-task notes. Each (method, env) cell uses $n = 3$ seeds (42, 123, 999), reported as mean $\pm$ SE with the t-distribution at $n-1=2$ d.f.; comparisons rely on non-overlapping SE intervals, a conservative choice at this sample size. The pivotal $0/6$ vs. $4/4$ contrast in Table 1 is decided by a Euclidean predicate, and Fisher's exact test rejects equal proportions at $p < 0.005$. The 12-episode pilot uses 4 episodes per env, with the VLM correctly identifying task-specific failure modes in every case; the $2/4$ PickAndPlace keyframe-agreement reflects heuristic under-localization rather than VLM error (closed-loop grip timing is the hardest mode for geometry-only oracles [20]).

E Broader Impacts and Reproducibility

This is methodological, simulation-only work; we make no real-robot or sim-to-real claims. On the positive side, fewer environment steps reduce GPU-hours, the IS-posterior framing offers a principled lens on VLM-in-replay, and the verified-CF gate is a transferable "language-prior + physics-test" pattern. On the negative side, the VLM-API energy and dollar costs are non-trivial (per-run spend is reported, and a heuristic Oracle fallback is available). Code, configs, prompt texts, the V-trace specialization, the two-regime q_ϕ proof, W&B run-IDs, and a VLM-RB [22] reproduction skeleton accompany the report.